

Pharmaceutical Report

Group 22

Executive Summary

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Background and Data

What is dementia and the importance of pharmaceutical treatment

Importance of equity across socioeconomic status

Pharmaceutical collection background

Note that dispensing doesn't immediately imply that the medication was taken

Association of dispensing between deprivation and time

Background on the NZDep2023 population counts by deprivation decile and health district, and the attributes included + suppression rules and unknown districts

The pharmaceutical data is a repeated cross sectional data frame showing how medicine was distributed in different regions over different years.

Note that just because medicine was dispensed does not mean that it was taken.

Additional district level variables to look into:

Income, age, employment, education, racial profile (Probably not), family dependencies.

Ethics, Privacy and Security

- Ethical considerations:

- Any comparison of groups being done in this project shouldn't cause have any implications about financial status between said groups. For example, Māori people take less medication than Pākehā because they can't afford them.
- Given the nature of the data collection, no causal conclusions can be drawn. The analysis will only be able to create commentary about the trends and relationships between the variables.

- Privacy considerations:

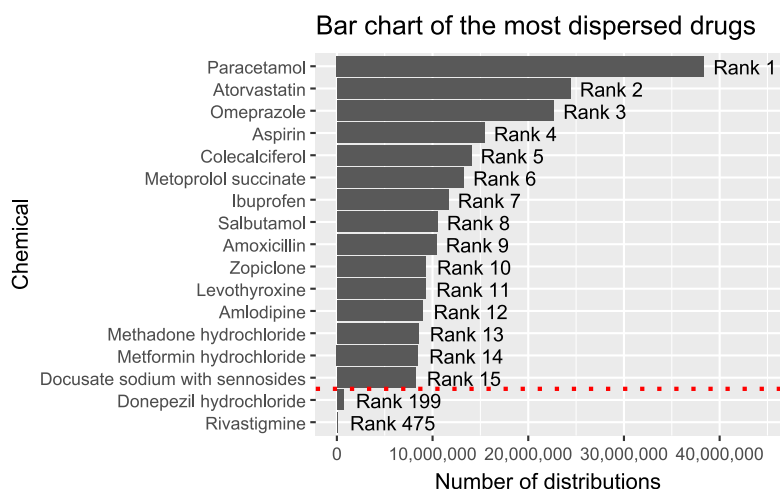
- We should not be removing the suppression layer from the count of medication distribution. While there is an oversight in how this data is provided that allows users to do this, and there

is code demonstrating that this is possible in this project, the values were randomly adjusted so that for the example instance doesn't have the true count revealed.

- For all analysis, the suppressed values will be imputed. Currently the method of replacing these with the median of the suppression range.

Exploratory Data Analysis

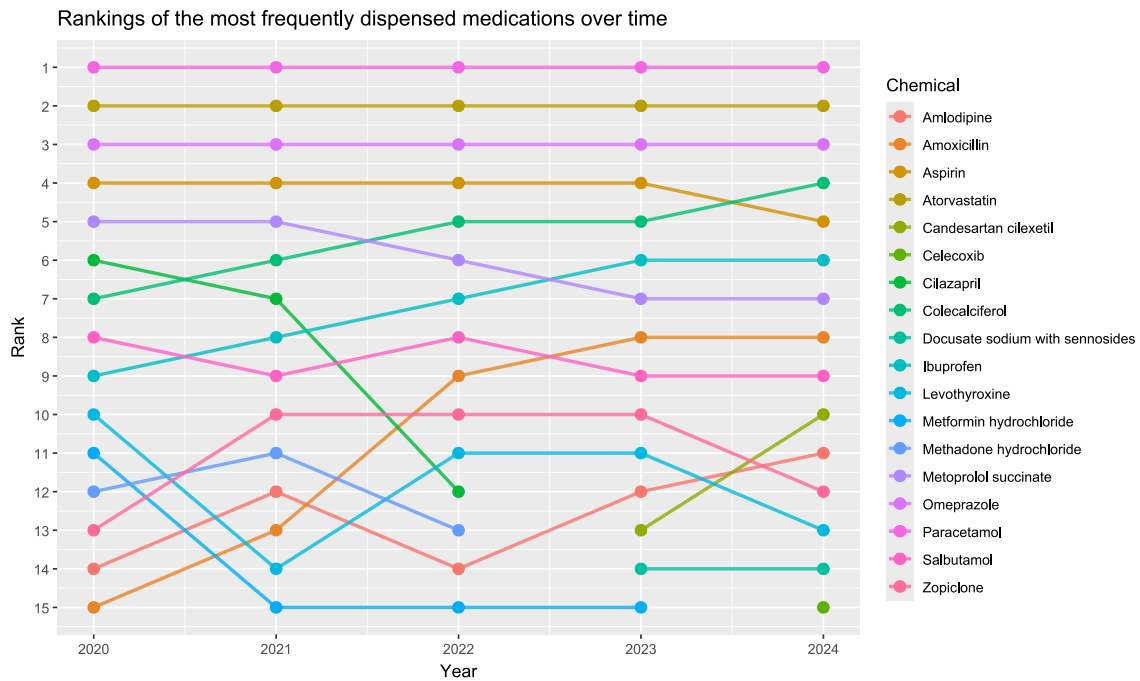
The graph below shows an unequal distribution of dispensing of medications of the 15 most widely distributed chemicals. Paracetamol is by far the most dispensed, with over double the amount of distributions compared to the aspirin, the chemical in 4th place. The second and third most widely distributed medications are Atorvastatin and Omeprazole. Overall, the distribution has a long tail. Donepezil hydrochloride and Rivastigmine aren't highly distributed enough to make the top 15, with the former ranking 199th across the 5 years of data, and the latter ranking 475th over the same period.



Looking at the line chart below showing rank over time, it can be seen that there are no changes to the rank of the top 3 medicines across the five years. Moving down the rankings, more variability is displayed. Colecalciferol rose from 7th to 4th, metoprolol succinate moved down from 5th to 7th and Ibuprofen from 9th to 6th. This relates to two things:

- The long tailed nature of the ranked distribution means that the steeper decreases in medication top of the ranks are rank unphased relative to any changes over time.
- The similar distributions over time between subsequently ranked medications are more likely to have their ranks changed by upward or downward trends in distribution over time.

This suggests that any future modelling would benefit from having year of distribution as a regressor, but it would be less likely to change the rankings of the highest distributed drugs



The table below shows the distribution count and ranks of the two dementia medications of interest. There has been an upwards trend in distribution for both of these medications, while rivastigmine has had its rank reduced in recent years. This is because while the distribution of this medication has increased in these years, not as much as other medication that had a lower distribution count that rivastigmine in 2021.

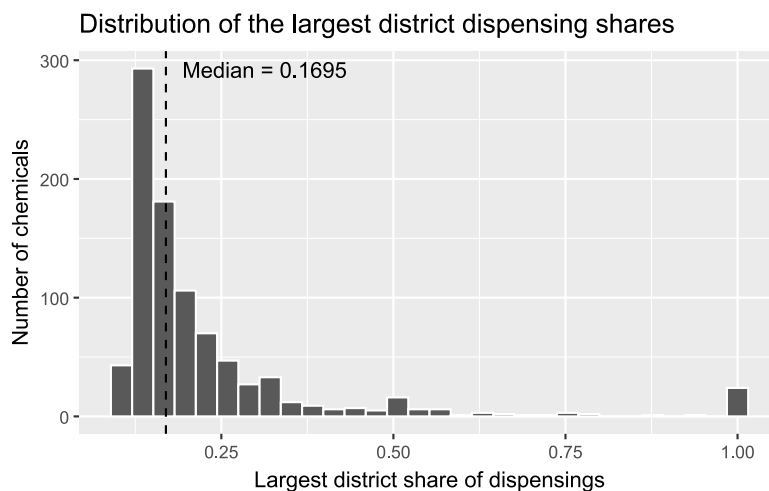
``summarise()`` has grouped output by 'YearDisp'. You can override using the ``.groups`` argument.

Year	Donepezil hydrochloride distribution count	Rivastigmine distribution count
2020	119502	6552
2021	121298	7545
2022	126223	7270
2023	137048	7688
2024	151711	8399

Donepezil hydrochloride distribution rank	Rivastigmine distribution rank
203	473

Donepezil hydrochloride distribution rank	Rivastigmine distribution rank
202	459
205	465
198	472
191	476

It is useful to know how frequently medications are distributed in a high concentration to a few health districts relative to an even spread across all health districts. One way to measure this is by using the proportion of distributions going toward the health district where most of the distributions occur. There are 19 health districts, and if the distributions occurred equally across said districts, this metric would take the value of 1/19. In the opposite extreme, if the medication was distributed in only 1 district, the metric would take the value of 1. Below is a histogram showing how each of the medications fall on this proportion scale.



The median value is 0.1695, meaning that a typical medication will have roughly 17% of their medication go towards the district that leads in distribution. The lowest observed proportion is 0.1049839 meaning that no medication is equally distributed across all districts. There's a small group of medications that have a proportion value of 1, meaning that they are only distributed in a single district. Note that this metric doesn't speak to the number of distributions for these medications, and are likely very niche with an estimated mean of 12.38 and max of 70 . The medications with these lower distributions are either very niche, available from retailers outside of pharmacies or not recorded in some districts.

Dementia as a risk is strongly associated with old age. This makes the districts with a higher count of aged population more relevant to this analysis than total population size in the health district. The graphs below show a positive relationship between aged population count and distributions of the two key dementia medications. This relationship is stronger for Donepezil Hydrochloride,

but both show that there are some observations above the line of best fit on the higher end of the district aged population. These observations belong to the {r}`dementia\_population` |> group_by(Chemical) |> slice_max(order_by = NumDisps, n = 5) |> pull(District) |> unique() district, and as the years increase it can be seen diverging further away from the line of best fit. Overall, this high correlation means that the older population size accounts for a large amount of the inter-district variation.

```
`geom_smooth()` using formula = 'y ~ x'
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Warning: The following aesthetics were dropped during statistical transformation: shape.

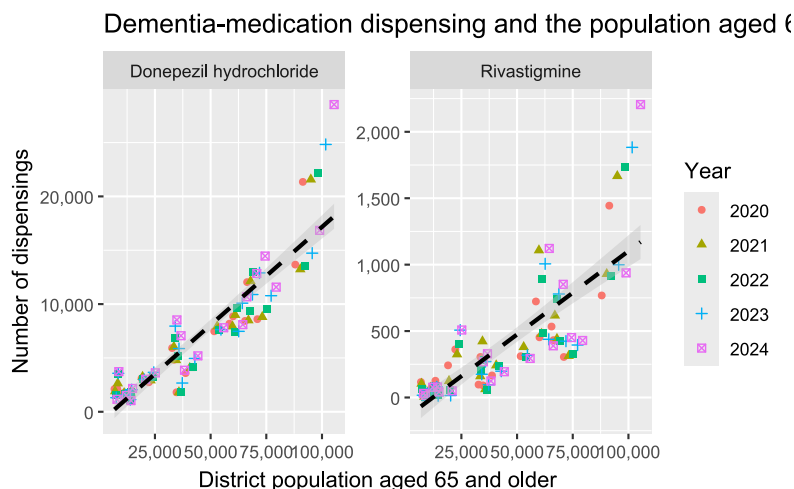
i This can happen when ggplot fails to infer the correct grouping structure in the data.

i Did you forget to specify a `group` aesthetic or to convert a numerical variable into a factor?

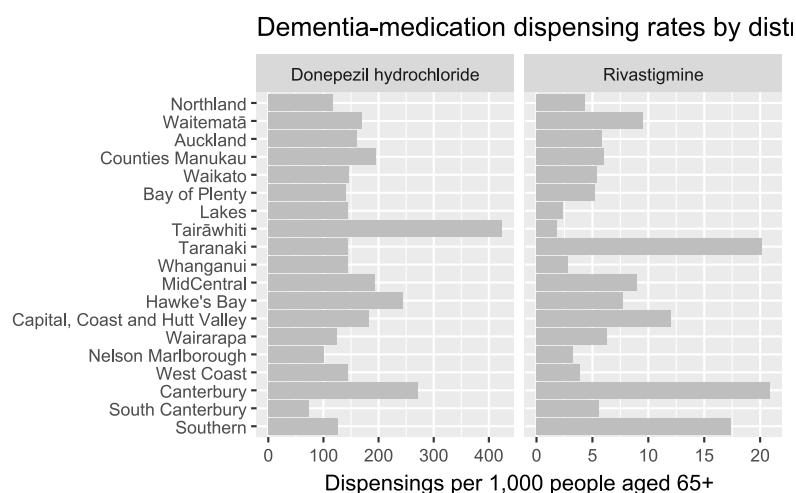
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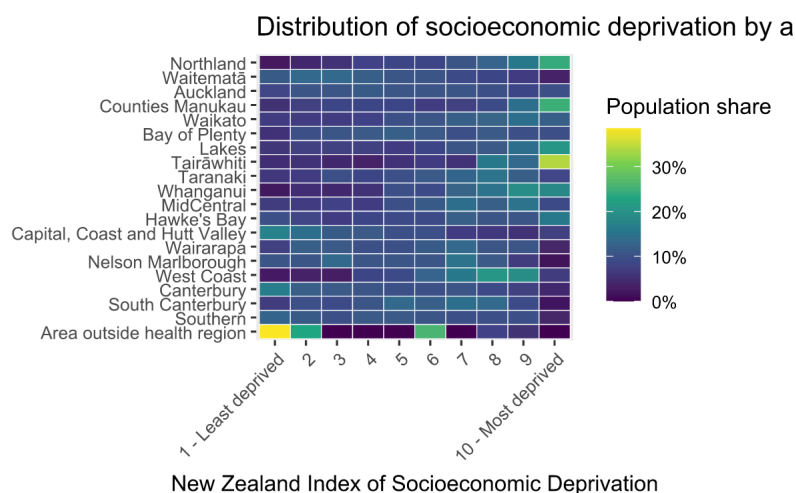
i Did you forget to specify a `group` aesthetic or to convert a numerical variable into a factor?



Crude rates were calculated as annual distributions per 1,000 people aged 65 years and over for the 2024 year. The table below shows the variation in these distribution rates across the different districts.



The first observation of note is that donepezil hydrochloride has a much higher rate than rivastigmine. Tairāwhiti has the highest overall rate when you take into account said larger rates for donepezil hydrochloride, as it is the district that has the highest rate for this medication. Canterbury and Hawke's Bay have the second and third highest rates respectively. Alternatively, Rivastigmine has the highest concentration of distribution in Canterbury, Taranaki and Southern. Taranaki doesn't stand out for being high or low with donepezil hydrochloride, and Tairāwhiti has the lowest distribution rate for rivastigmine. This could be related to the high proportion of the population of Tairāwhiti falling higher on the latest (2023) NZ deprivation scale as seen below.



Detailed Analysis Results

Potential models:

Count data suggests either Poisson model, or negative binomial if overdispersion needs to be accounted for (Create both and compare the diagnostics of the two)

Inclusion of interactions. Why they were/weren't required

Include confidence intervals for the different coefficients

Limitations:

Only 2023 census returns are informing the deprivation profiles for the health districts, whereas the pharmaceutical dispensings are annually temporal

All data is summarized to the district level, so we can only look at trends in a region as opposed to any individual level information

Conclusions and Recommendations